Object Oriented Design Patterns

Kevin Ceausu B00128171

Assignment – Study Planner

1. Application Overview

The application developed for this assignment is a Study Planner and Assignment Tracker application built using Java. My application allows users to organise their work for school or university through projects and tasks, having the option to set deadlines and priority for each.

The system provides a graphical user interface where users can create projects, assign tasks, set deadlines for them and mark them as complete once done. The application automatically updates once changes occur to the tasks, providing real time feedback to users.

The application was developed using multiple object oriented design patterns to improve flexibility, scalability and be easy to maintain. These include Builder, Composite, Observer and Decorator patterns.

2. Functional Requirements

The system must:

- Allow users to create projects
- Allow users to create tasks within projects
- Allow users to delete projects and tasks
- Allow users to mark tasks as complete
- Display the projects and tasks in a hierarchical structure
- Display information available about selected tasks
- Automatically update interface when data changes
- Display total estimated hours
- Display total and completed task counts

3. User Requirements / User Interface

The system provides a graphical user interface that consists of:

- A tree panel displaying projects and tasks in a hierarchical structure
-A detail panel displaying information about the selected task
- A toolbar containing buttons for adding projects, tasks, deleting them or marking them as complete
- A status panel displaying total hours, total tasks and completed tasks.

Users interact with the system by selecting items from the tree and using the toolbar buttons to modify the data. When changes are made, the interface automatically updates.

4. Design Patterns Explanation

A. Builder Pattern

The builder pattern is used in the task class to construct task objects. A task contains multiple fields such as title, description, due date, priority, estimated hours and completion status. Instead of using a large constructor, the builder pattern allows step by step construction of a task object, helping a lot with readability and flexibility of the project.

I chose the Builder pattern over traditional constructors because it allows clear and maintainable object creation.

B. Composite Pattern

The composite pattern is used to represent the hierarchical structure of projects and tasks. The PlannerComponent class acts as the base component, while TaskLeaf represents tasks and ProjectComposite represents projects that can contain other components. By using the methodology, my system is able to treat individual tasks and projects uniformly, making operations such as calculating the estimated hours simple.

I chose the Composite Pattern because the project involves a tree structure where objects can contain other objects.

C. Observer Pattern

The Observer pattern is used to update the interface automatically when changes occur. The PlannerModel acts as the subject while components such as TaskTreePanel and StatusPanel act as observers. When the model is changed like a new task is added or deleted, all observers are notified and update their display.

I chose the Observer pattern as it ensures the user interface remains consistent with handling data without requiring manual updates.

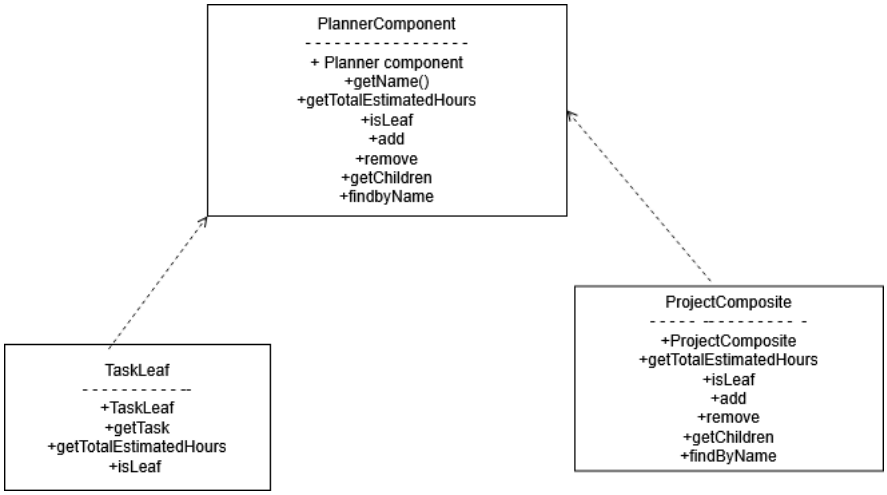
D. Decorator Pattern

The decorator Pattern is used to dynamically modify the appearance of the tasks. The TaskView interface defines the base component, while BasicTaskView provides the default implementations. Decorators such as HighPriorityDecorator, CompletedDecorator, OverdueDecorator wrap the base view to add additional visual behaviour.

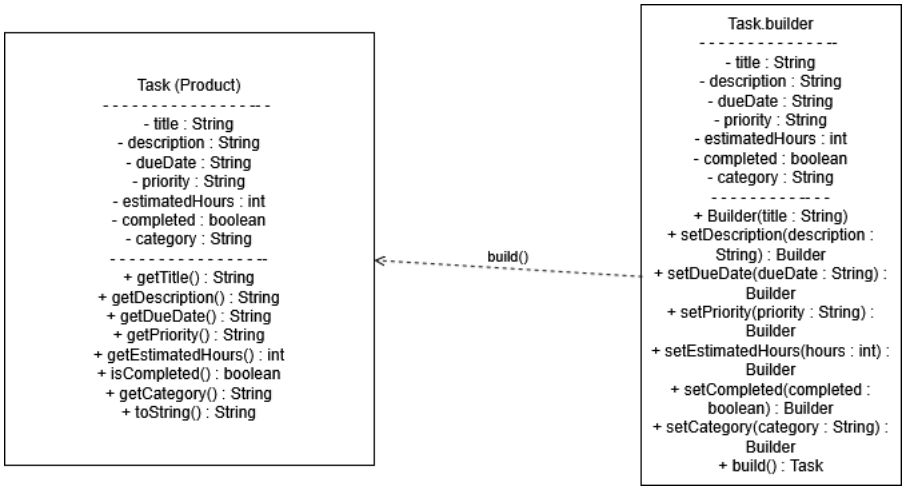
I chose the Decorator Pattern because it allows me to add multiple visual features and combine them without creating many subclasses.

5. UML Diagrams

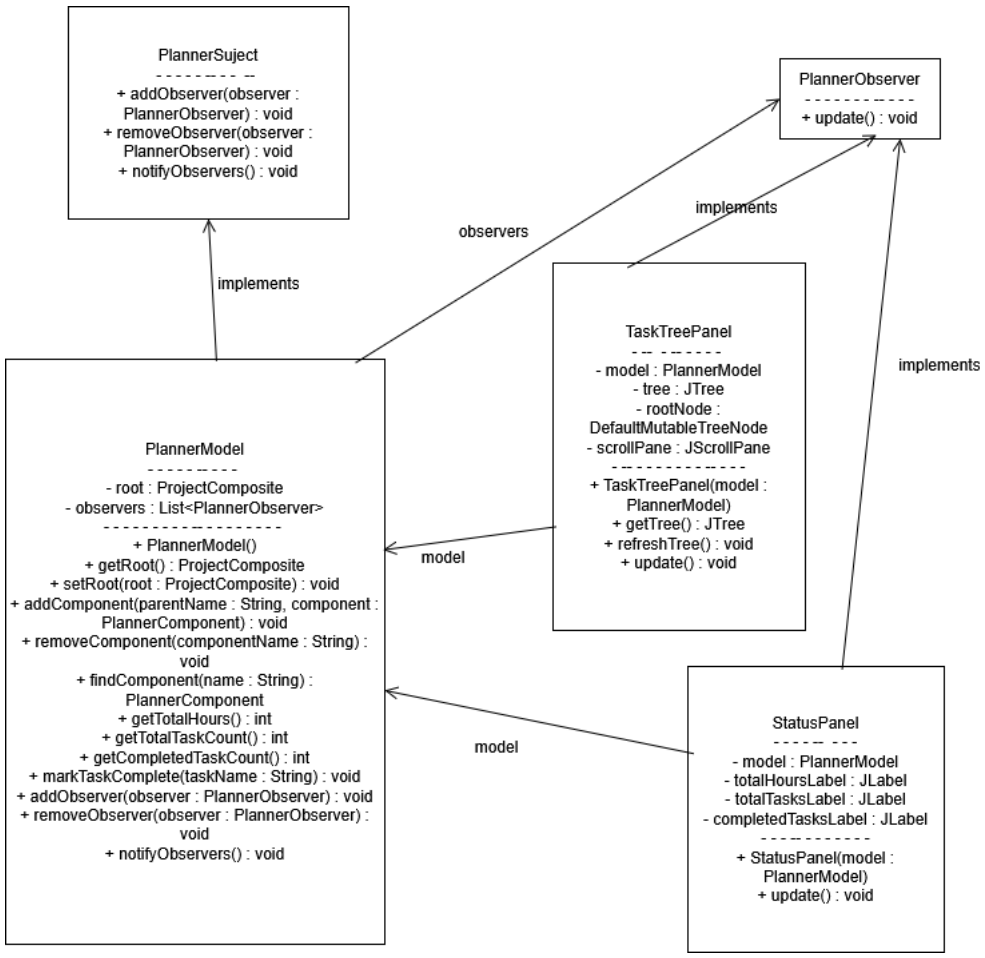
Composite UML



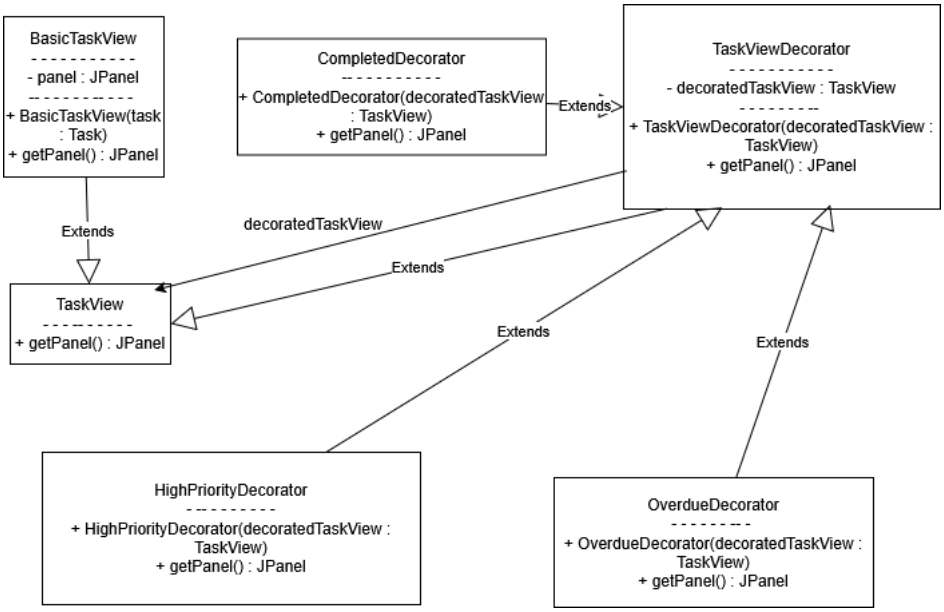
Builder UML



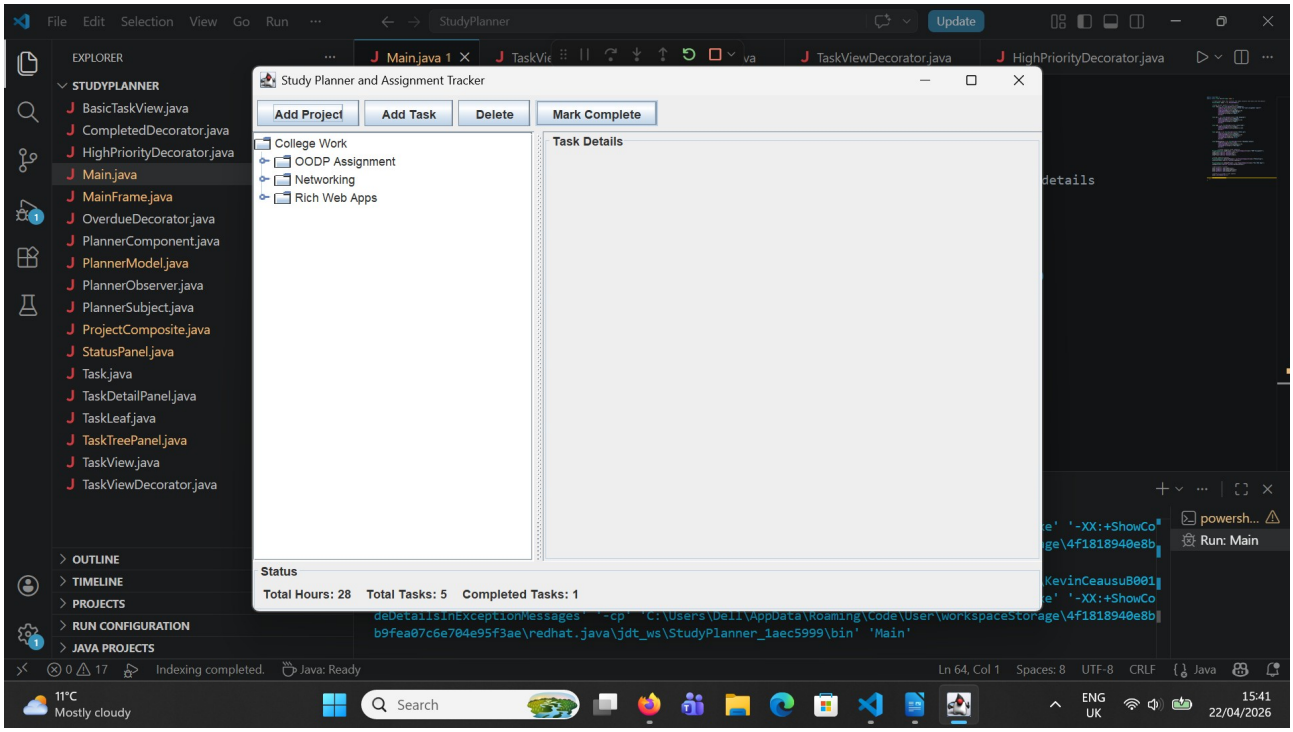
Observer UML

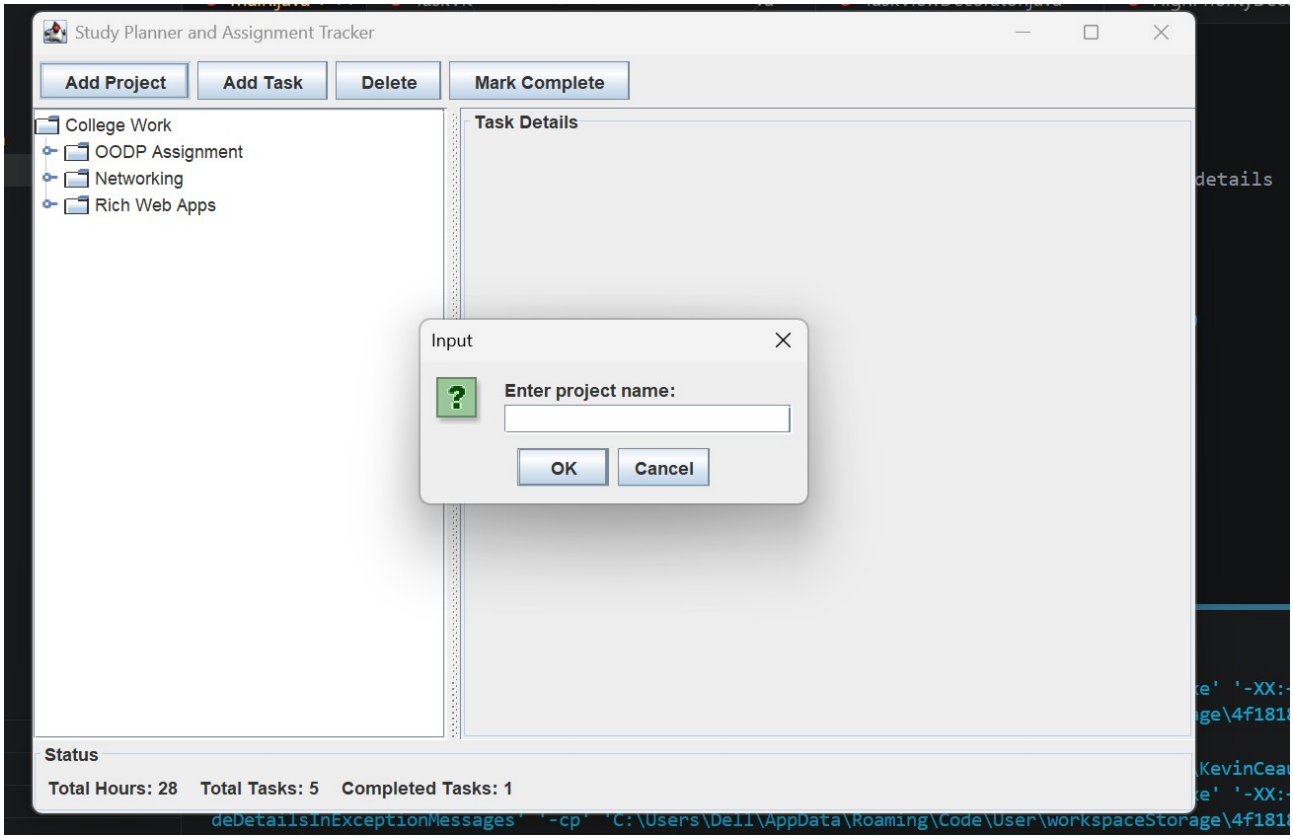


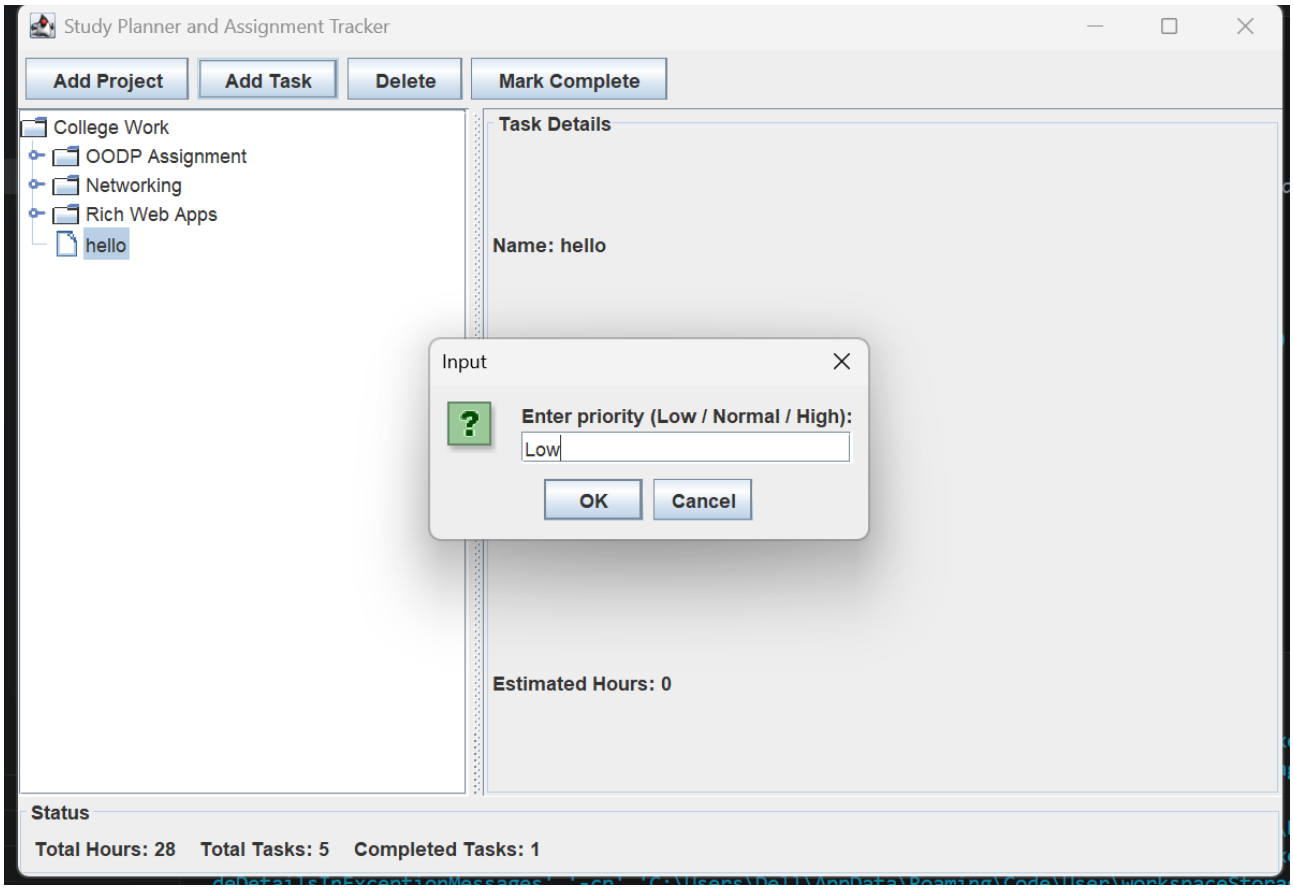
Decorator UML

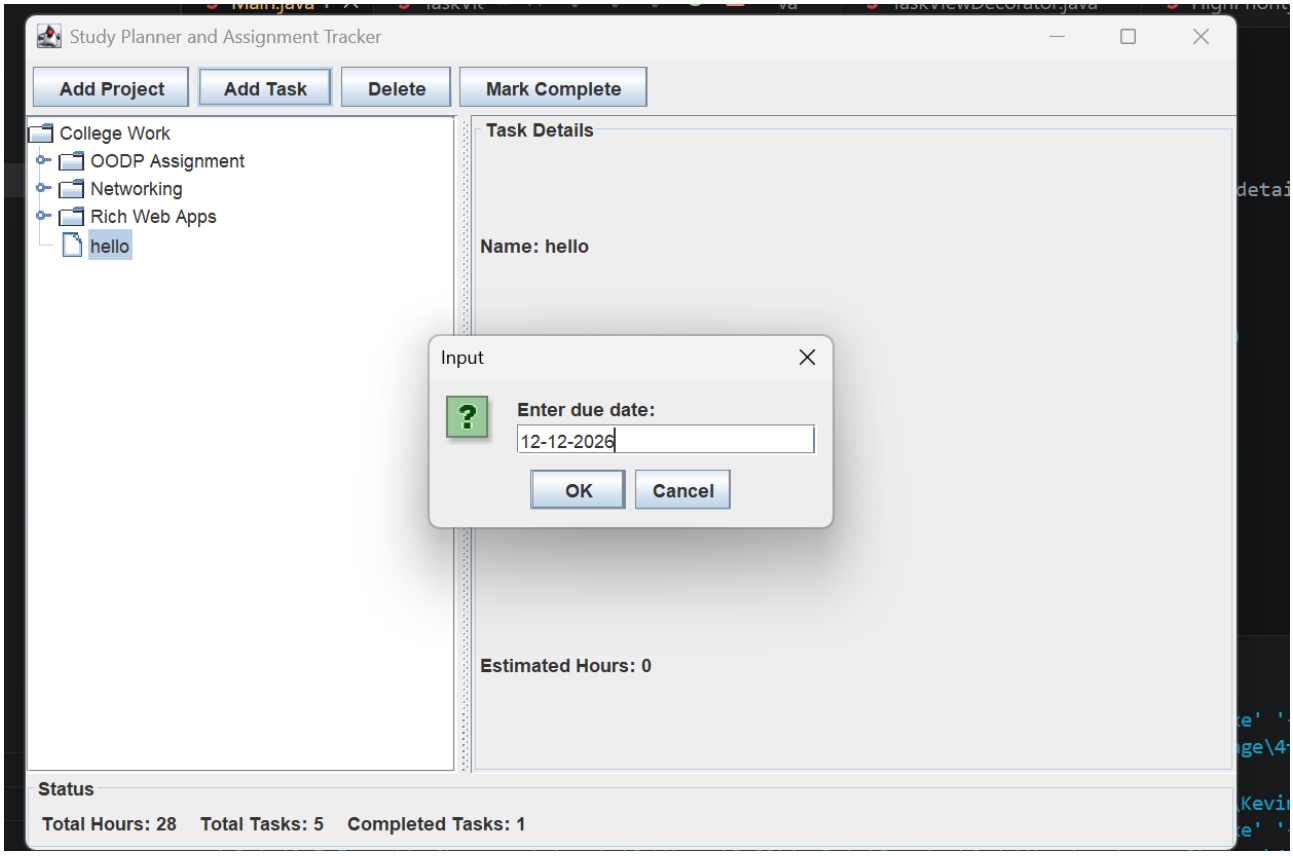


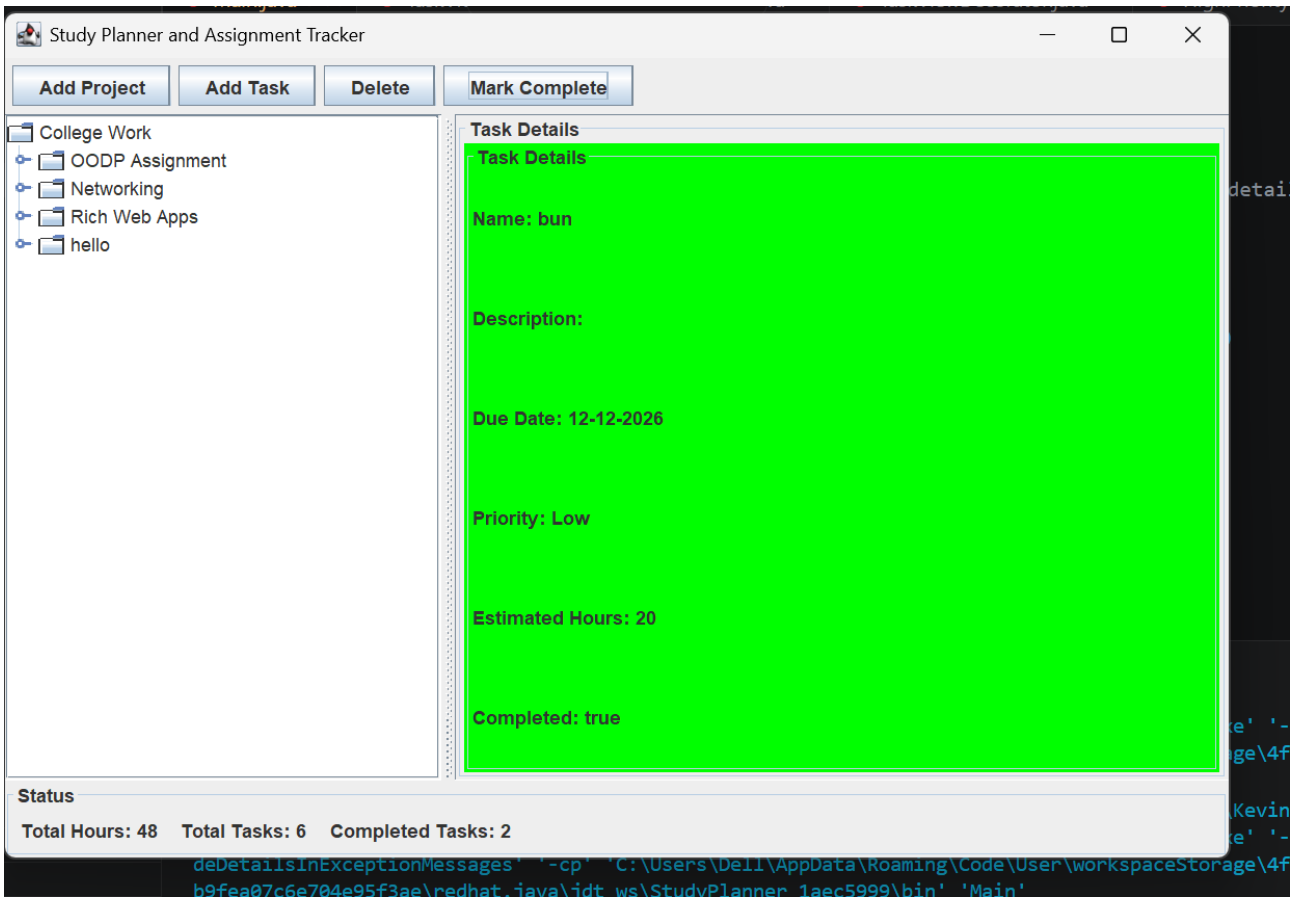
6. Screenshots of Application on Execution




Study Planner and Assignment Tracker
Add Project
Add Task
Delete
Mark Complete
College Work
OODP Assignment
Networking
Rich Web Apps
Task Details
Input
Enter project name:
OK
Cancel
Status
Total Hours: 28
Total Tasks: 5
Completed Tasks: 1


Study Planner and Assignment Tracker
Add Project
Add Task
Delete
Mark Complete
College Work
OODP Assignment
Networking
Rich Web Apps
hello
Task Details
Name: hello
Input
Enter priority (Low / Normal / High):
Low
OK
Cancel
Estimated Hours: 0
Status
Total Hours: 28
Total Tasks: 5
Completed Tasks: 1


Study Planner and Assignment Tracker
Add Project
Add Task
Delete
Mark Complete
College Work
OODP Assignment
Networking
Rich Web Apps
hello
Task Details
Name: hello
Input
Enter due date:
12-12-2026
OK
Cancel
Estimated Hours: 0
Status
Total Hours: 28
Total Tasks: 5
Completed Tasks: 1


Study Planner and Assignment Tracker
Add Project
Add Task
Delete
Mark Complete
College Work
OODP Assignment
Networking
Rich Web Apps
hello
Task Details
Task Details
Name: bun
Description:
Due Date: 12-12-2026
Priority: Low
Estimated Hours: 20
Completed: true
Status
Total Hours: 48
Total Tasks: 6
Completed Tasks: 2